

FROZEN PHASE-10R MODEL
MultiScale PointNet2 + TargetQuery Transformer
117 Target Ontology | $S_{\text{global}} = 500 \text{ mm}$

BRANCH 11A: AMOS22 COHORT
600 Clinical Subjects (500 CT, 100 MRI)
Independent External Medical Center

External Surface Extraction (Organ Mask Blind)
Intensity-based CT / Border Otsu MRI
4096 XYZ Normalized Points

EVALUATION: CLINICAL ANATOMICAL GT
True Internal Organ Centroids (15 Targets)
Macro/Micro MRE, Median, SDR@5-40
Zero-Shot Cross-Modality & Cross-Dataset

BRANCH 11B: HuMMan REAL SENSOR
Real iPhone Depth + Camera Calibration
Independent Physical Depth Sensor

Real Depth Back-Projection ($u, v, Z \rightarrow XYZ \text{ mm}$)
Sensor-Only H1 & SMPL-Assisted H2 Filtering
Canonical Anatomical Frame Alignment

EVALUATION: SENSOR REALISM (ZERO GT)
NO ORGAN GROUND TRUTH (NO MRE)
Model Acceptance (NaN=0%, Out-of-Body)
Temporal Jitter $J_k(t)$ & Cross-View Disagreement

BRIDGE ANALYSIS

Simulated vs Real Sensor Loop